Design Verification Plan & Report - VBF-T2R2-DVPR-01

Case t2_r2 - Virtual Battery Factory - generated 2026-08-26

Design Verification Plan & Report (virtual, one row per item; determinations vs task criteria)

Item	Condition	Result	Determination	Source
1C discharge capacity	25 C amb, DFN, 1C CC to 2.5 V	6.9089 Ah	reference (no task threshold; nominal 5.0 Ah)	final_1c_dfn.json
4C temp. rise	45 C amb, lumped thermal	T_max 360.089 K = +41.939 K (86.9 degC)	no task red-line; recorded (see DFMEA)	final_4c_dfn.json
4C plating	plating on; anode pot. < 0 V = plating	min +0.04095 V	PASS (task criterion)	final_4c_dfn.json
Voltage window	parameter set upper/lower cut-offs	2.5 - 4.2 V	recorded (set baseline)	chen2020_dump.json
SEI @ 100 cyc (1C)	aging 1C, SPMe, ec-reaction-limited	99.946 nm	PASS vs <= 500 nm	final_aging100.json
SEI @ 500 cyc (1C)	--cycles 500	330.138 nm	PASS vs <= 550 nm	final_aging500.json + derived key bridge
-20 C retention (lumped)	lowT 1C, starts 298.15 K	99.6205 %	PASS vs >= 90 %	derived/final_retention.json
-20 C retention (cold soak)	isothermal 253.15 K	99.4300 %	PASS backup caliber	derived/final_retention_isothermal.json
Energy density	calc-energy on 1C DFN	522.4524 Wh/kg	PASS vs >= 327.18 Wh/kg	final_calc.json
Nail / overcharge-TR / crush / drop / rate-pulse IR / EOL cycle count	-	-	N/A (beyond simulation boundary, requires physical experiment)	-

Conclusion

Task criteria: 5/5 PASS (ED, SEI@100, SEI@500, -20 C retention, 4C no-plating). Reference protocol items (1C capacity, 4C temp rise, voltage window) recorded - no task thresholds given for them. Uncovered items explicitly N/A above (paper limitations). Honest caveats: -20 C lumped warm-start (isothermal backup 99.430 % PASSES); 4C acceptance 0.958 Ah (~13.9 %) and T_max 86.9 degC recorded; SEI values rely on coating-bridge k = 2.00e-15 m/s (estimate).